

User Manual

IDA Indoor Climate and Energy

Version 4.8

EQUA Simulation AB

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1. About the Manual

This manual gives a general introduction to IDA Indoor Climate and Energy (IDA ICE).

Before or in parallel with reading the manual, it is advisable to follow the Getting Started Guide, which is a separate document. This will bring you through a worked example and fill in supporting information in the process.

The documentation for individual forms, dialog boxes and reports is available as on-line help texts. Pressing F1 on your keyboard while a form or dialog is active will generally open the appropriate topic. Access to the on-line help texts, tutorial movies etc. is also available from the Help menu. From version 4.7, a Portal page is also opened when the application is started with various links to useful information.

Another key source of information, documentation, sample models, scripts etc. is the IDA ICE Forum, an interactive web site where users can ask questions, which are then answered by the support team and/or other users. The forum is available from the Help menu and the Portal page.

Chapters 2 and 3 provide a general overview of program management and simulation models. Chapter 4 treats import of CAD data and chapter 5 is an introduction to working at the advanced level. In Chapter 6 some tips for getting optimal performance are given.

2. Basic principles of IDA and IDA Indoor Climate and Energy

2.1. Introduction

IDA Indoor Climate and Energy (IDA ICE) is a program for study of indoor climate of individual zones within a building, as well as the energy consumption for the entire building. IDA ICE is an extension of the general IDA Simulation Environment. This means that the advanced user can, in principle, simulate any system whatsoever with the aid of the general functionality in the IDA environment.

Normally, the system to be simulated consists of a building with one or more thermal zones, a primary system (plant) and one or more air handling systems. Surrounding buildings might shade the building. The air inside the building contains both humidity and carbon dioxide. Weather data is supplied by weather data files, or is artificially created by a model for a given 24-hour period. Consideration of wind and temperature driven airflow can be taken by a bulk air flow model. Predefined building components and other parameter objects can be loaded from a database.

2.2. The three levels of user interface

The user interface is divided into three different levels, with different support and scope for the user. At the simplest level, called wizard, the scope is limited to a certain type of study and level of approximation. The user is given the opportunity of carrying out a simulation directly, or transferring the data entered to the next level, called the standard level. From version 4.7, a new wizard interface IDA Early Stage Building Optimization (IDA ESBO) is included, see Figure 2.1. It replaces the IDA Room wizard which is no longer included. To learn more about IDA ESBO, press  on the toolbar and then the F1 key on your keyboard to launch the online help.

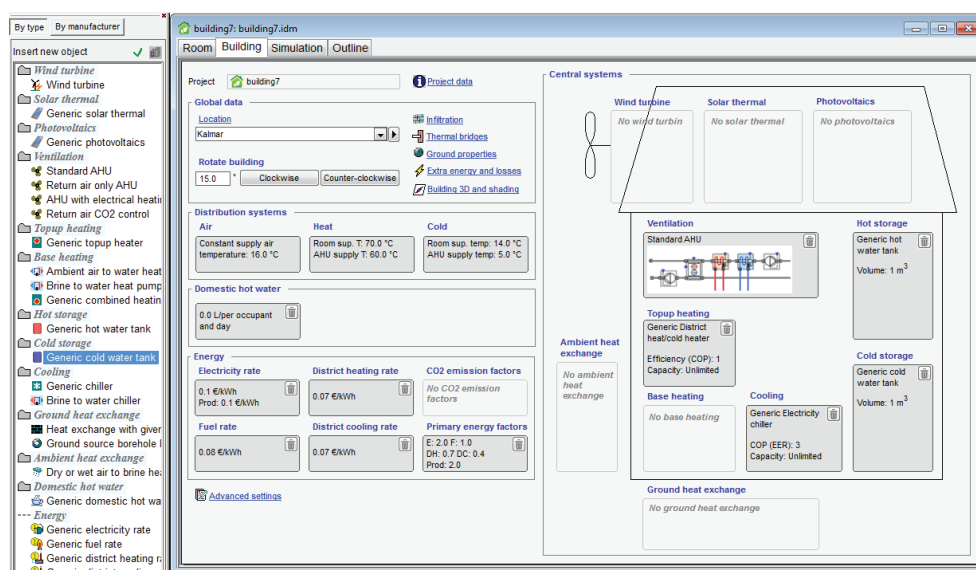


Figure 2.1 The Building tab of IDA ESBO

At the standard level (Figure 2.2), the user is given greater freedom to design a building model. This level defines geometry, materials, controller settings, loads, etc., in a manner that should be easy to handle for a majority of engineers. The basic steps of using the standard level are covered by the Getting Started Guide. An interactive Process guide (Help menu) is also available to guide you, by movies and other support, through the steps of building a model at the standard level.

Name	Group	Floor height, m	Room height, m	Floor area, m ²	Heat setp., °C	Cool setp., °C	AHU	System	Supply air, L/s, m ²	Return air, L/s, m ²	Occup. no./m ²	Lights, W/m ²	Lights, kWh/m ²	Equipment, W/m ²	Equipment, kWh/m ²	Ext. win. area, m ²	Occup. schedule	Light schedule	Equip. schedule
Totalm2		0.0	2.6	10.0	21	25	Air Ha...	CAV	2.0	2.0	0.1	5.0	43.8	15.0	131.4	1.5	@Alwa...	@Alwa...	@Alwa...

Figure 2.2 Main form for the building at standard level

At the advanced level (Figure 2.3), the simulation model is no longer defined in physical terms, but in the form of connected component models, defined by equations. At this level, the individual time evolution of variables can be studied. All equations, parameters and variables can be examined at this level.

A user of the Expert edition of the program may also edit the connection structure at the advanced level. Some of these operations are easy to carry out, e.g. changing a proportional controller to a thermostat. Others are more complicated and require a deeper knowledge of the design of the models.

Use of the advanced level is introduced in Chapter 5. EQUA also maintains some exercises that can be used to gain familiarity with the advanced level. In addition, a great deal of information is found in the on-line help texts. Make sure also to visit the User Forum, which can be accessed as soon as you have IDA ICE running. You start the User Forum either from the Portal page (the first screen that meets you) or from the Help menu -> Support-> Ask a question.

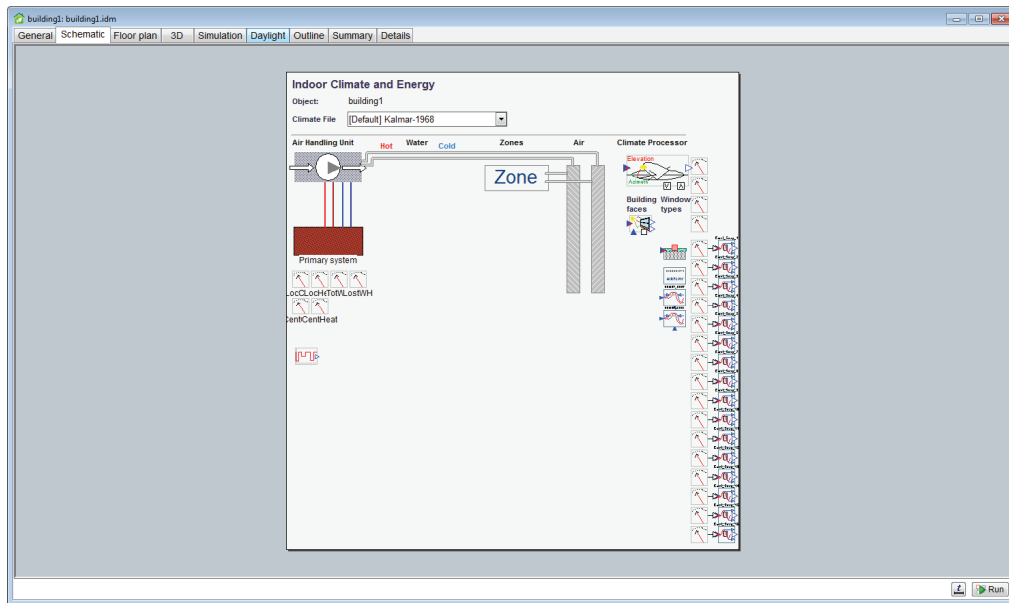


Figure 2.3 Main Form for the building at the advanced level

2.3. Forms and dialogs

IDA ICE is built up around forms and dialogs. The forms contain no Cancel button, i.e. there is no access to earlier versions of a form (except by using Undo). Forms do not lock each other, and several windows with forms can be open simultaneously (but only the one where work is in progress is active). A form can be printed. Simulations and many other operations can be carried out without having to first close open forms. Dialogs, i.e. input windows with OK and Cancel buttons, work in IDA as in most other Windows programs. They lock everything else but the current dialog window.

3. Model description

This chapter treats the mathematical models of IDA ICE together with some of their input forms. The reader is assumed to have mastered the basics of the program by first following the getting started guides.

A building model consists of a single or several thermal zones, optional central air handling units and a single primary system. When a new model is initiated at the standard level, a default air handling unit and primary system are normally automatically inserted (may depend on localization). The default systems have unlimited capacity for providing the zones with air and water at given temperatures. By default, the supply air temperature is kept constant at 17°C; the chilled water temperature to zones is 15 °C and the heated water temperature is a function of the outdoor air temperature. For many studies nothing needs to be altered in the HVAC systems.

The default air handling unit (AHU) can be removed, but a plant object must always be present in a model. (However, without an AHU or any water based room units, the plant will not use any energy.)

This description deals firstly with the primary system (plant), then follows the supply chain to the AHU (Air Handling Unit), and finally to the zones.

3.1. The Primary system (Plant)

The primary system may be described in two separate ways, by using the Standard plant or by replacing this by an ESBO plant model.

3.1.1. The Standard plant

In the default configuration, the primary system consists of six components (designated 1-6 respectively in Figure 3.1): chiller (1) and a schedule (2) for its operation as well as a boiler (3), a controller (4) for hot water supply temperature and a schedule (5) for night setback operation. Also connected to the boiler is a schedule for its operation (6). The six energy meters in the lower right corner monitor energy consumption of various categories in the primary system.

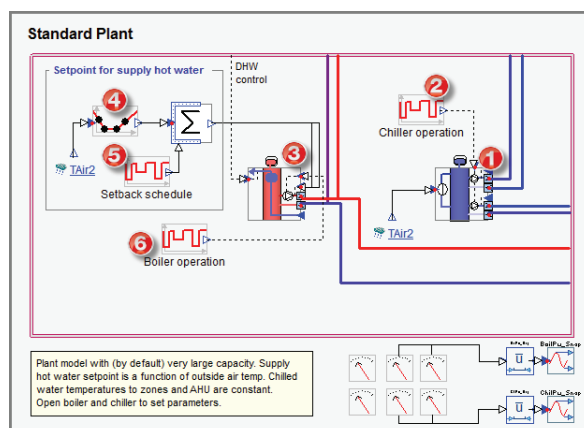


Figure 3.1 The primary system in the default configuration

3.1.1.1. The Boiler

The boiler converts purchased energy, e.g. gas, electricity or district heat, to warm water with given temperature and pressure for circulation through water based heat exchangers in the building. It also uses energy for production of domestic hot water and pumping. Boiler efficiency is by default constant, as specified in the Defaults form. Open the component to view key parameters.

It is also possible in the standard boiler to model water heating efficiency as a function of boiler temperature and part load. The model and parameter definitions used are the same as those of EnergyPlus. Currently, there is no database support for this performance data since the IDA ESBO interface offers alternative ways of modeling more realistic equipment.

Pumping power consumption for heating water circulation can be specified in three ways: (1) proportional to the water flow through the boiler (default), (2) as a proportion of distributed heat, or (3) as a polynomial function of the water flow. The third option follows the conventions of ASHRAE 90.1

The first option assumes an ideal pressure controlled pump with constant efficiency. Alternatively, by setting the efficiency to some large number and specifying the k1 parameter, pumping effort can be given in proportion to distributed heat. (PSetMax should still have a somewhat realistic value.)

In the third option, the efficiency of different types of pumping solutions and control are reflected by a user provided polynomial function. This requires the input of a design massflow as a point of reference. Two standard curves from ASHRAE 90.1 are provided.

Pumping power for domestic hot water circulation can also be specified via k2, which specifies pumping power as a fraction of domestic hot water heating power.

The setpoint for the hot water supply temperature comes from a special controller component connected to the boiler. The controller provides a graph showing the setpoint as a function of the outdoor air temperature. (Press F1 in the dialog for detailed instructions.)

3.1.1.2. The Chiller

The chiller and its circulation circuit operate in a similar way, but differ in some aspects from the boiler. The chiller uses electrical power to produce chilled water at two different constant temperatures (but with the same pressure). The colder water, normally 5 °C, supplies the AHU. The somewhat warmer temperature, normally 15 °C, goes to the zones.

Similarly as for the boiler, EnergyPlus correlations for modeling temperature and part load behavior may be applied. Pumping power is also specified as in the boiler.

3.1.2. The ESBO Plant

By replacing the default Standard plant with an ESBO Plant, more complex plant models may easily be constructed.

In the General tab, many different system configurations can be described by dragging in various components from the palette (c.f. Figure 3.2). See the ESBO documentation for more details about available components.

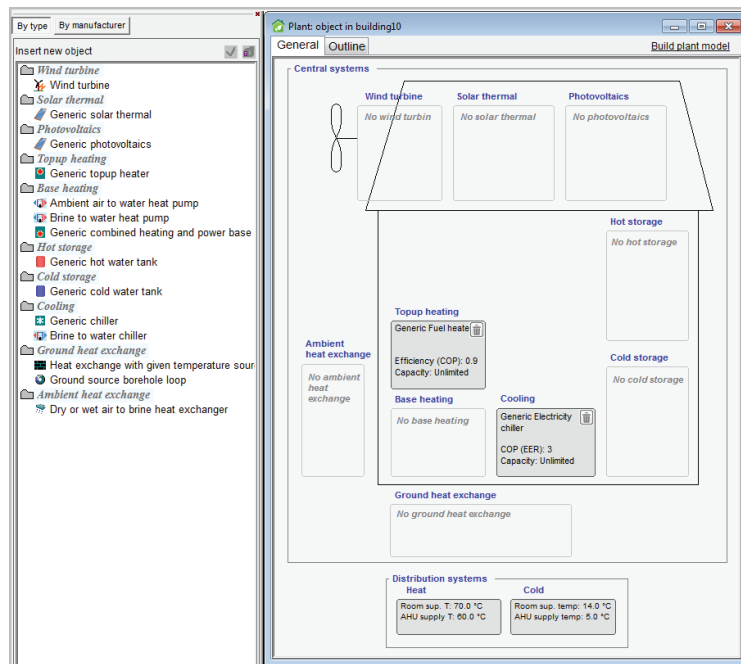


Figure 3.2 The ESBO Plant – General view

In the next few sections, we describe the principles for the configuration and control of the default ESBO systems. Here, the narrative will rely on system schematic diagrams as displayed in the Advanced level. Press Build plant model in order to generate the Advanced level for the ESBO Plant.

3.1.2.1. Purpose and ambition of the default generated systems

In economically feasible real systems many compromises must be made. The total number of valves, pumps and connections must be reasonable and control logic must not be overly complex. In contrast, the purpose of the Esbo default systems is to exploit all significant energy flows based on a minimum of input data from the user even if the physical components needed are not economically justifiable in a real system. The generated systems are technically realistic but might, at least for smaller installations, contain too many parts to be economically attractive for realization in real life.

Furthermore, the control systems employed are extraordinarily “decentralized.” In order to avoid the combinatorial explosion of having to define a central control system for each possible system configuration, each subsystem is maximally self-sufficient. A flow controller for, e.g., a ground source circuit will simply check when it is possible to deliver heat (or cold) and does this until it is not energy-wise economical to continue (when pumping costs will exceed the benefits). Long-term control strategies, for example to try to seasonally balance a large thermal store, is out of scope of the automatically generated default systems. Whenever the short-term benefit of a certain flow is in place, the flow will be activated.

3.1.2.2. The default tank models

All generated systems are centered around two stratified water tanks, one for hot and one for cold water. These two tanks are mandatory for most system combinations, also for situations

where a real system would not include a tank. Furthermore, the tanks are by default equipped with valves that will allow each entering and exiting flow to be made at the optimal height in the tank. This way, minimal buoyant mixing will occur. Real tanks exist that behave in a similar fashion, using for example heat lance technology. The tank is also by default equipped with a bypass (shunt) circuit for each client-side (discharge) connection. This way, return water from the same circuit is used first to attain the target temperature of the required supply flow, thus minimizing the flow through the tank.

For situations where a storage tank is not part of the concept to be investigated, the volume of the mandatory tanks is set to be fairly small with respect to surrounding systems, but not to zero. The tanks can in this situation be thought of as “temperature switchboards” that minimize the amount of temperature mixing in the system. Tank volume would then represent the thermal mass of the actual HVAC system.

The default height to diameter ratio (shape factor) of the tanks is set to 5, representing a fairly tall tank with, correspondingly, only a small amount of conduction between the water layers. In order to create a less ideal tank and ultimately approaching a completely mixed tank, the shape factor can be set smaller, so that conduction will even out temperature differences between layers. Another way of gradually approaching a well-mixed tank is to decrease the number of layers.

When working at the Advanced level, it is possible to turn off the ideal heat lance feature of the tanks and select fixed vertical positions for tank ports. Buoyant mixing will then occur if a hot water stream enters the lower part of a tank. The IDEAL parameter in the tank controls the heat lance feature. Discharge-side bypass flow is controlled by the SHUNT parameter. For a full mathematical description of the tank model, c.f. the Forum post <http://forum.equa.se/question/2777/stratified-tank-model-equations/> (open forum from IDA ICE before following the link).

3.1.2.3. An example of a generated system

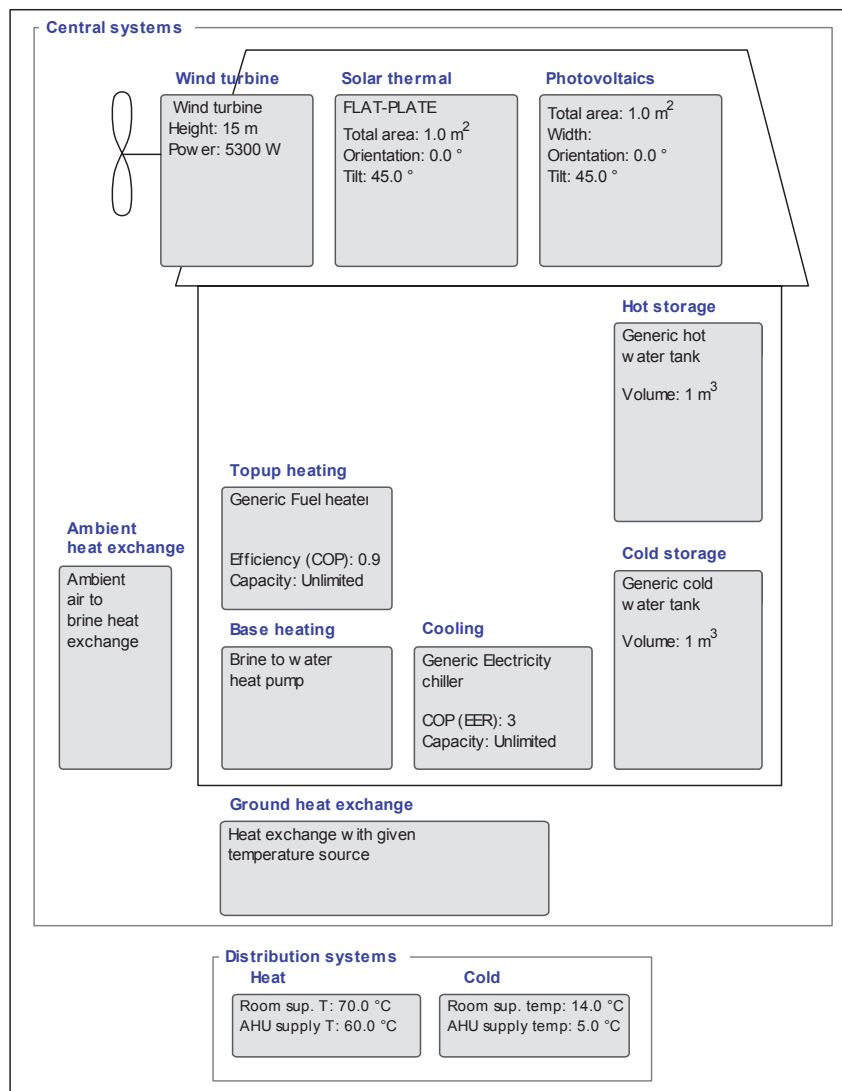


Figure 3.3 shows an example of how the General tab may be populated with models. Some model has been chosen for each of the possible functions. Brine to water vapor compression machines have been defined for base heating as well as for cooling. Both ground heat exchange and ambient heat exchange have been defined. Below is a relatively detailed account of the workings of the plant model for this case. It can safely be omitted on a first reading, as no references are made to it in the subsequent text.

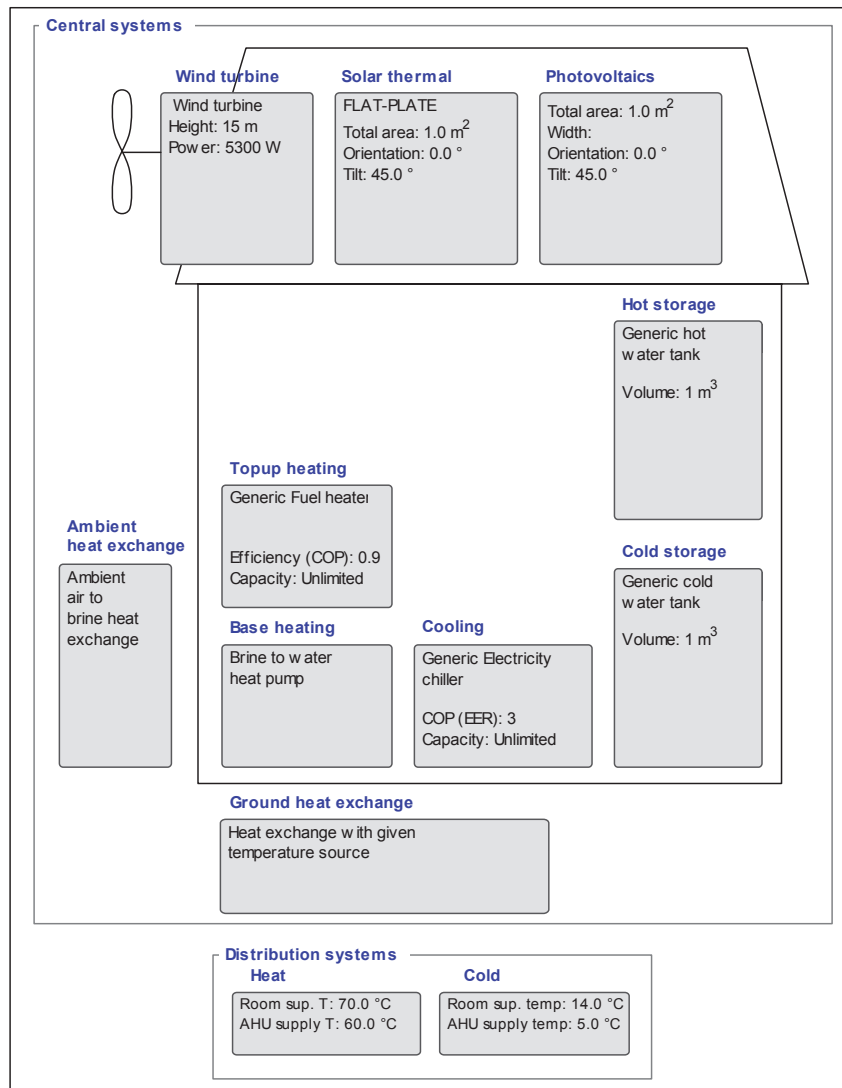


Figure 3.3 An example of the General tab of the ESBO Plant populated with models

Figure 3.4 shows the generated system at the Advanced level of IDA ICE. We will go through and explain each major subsystem below.

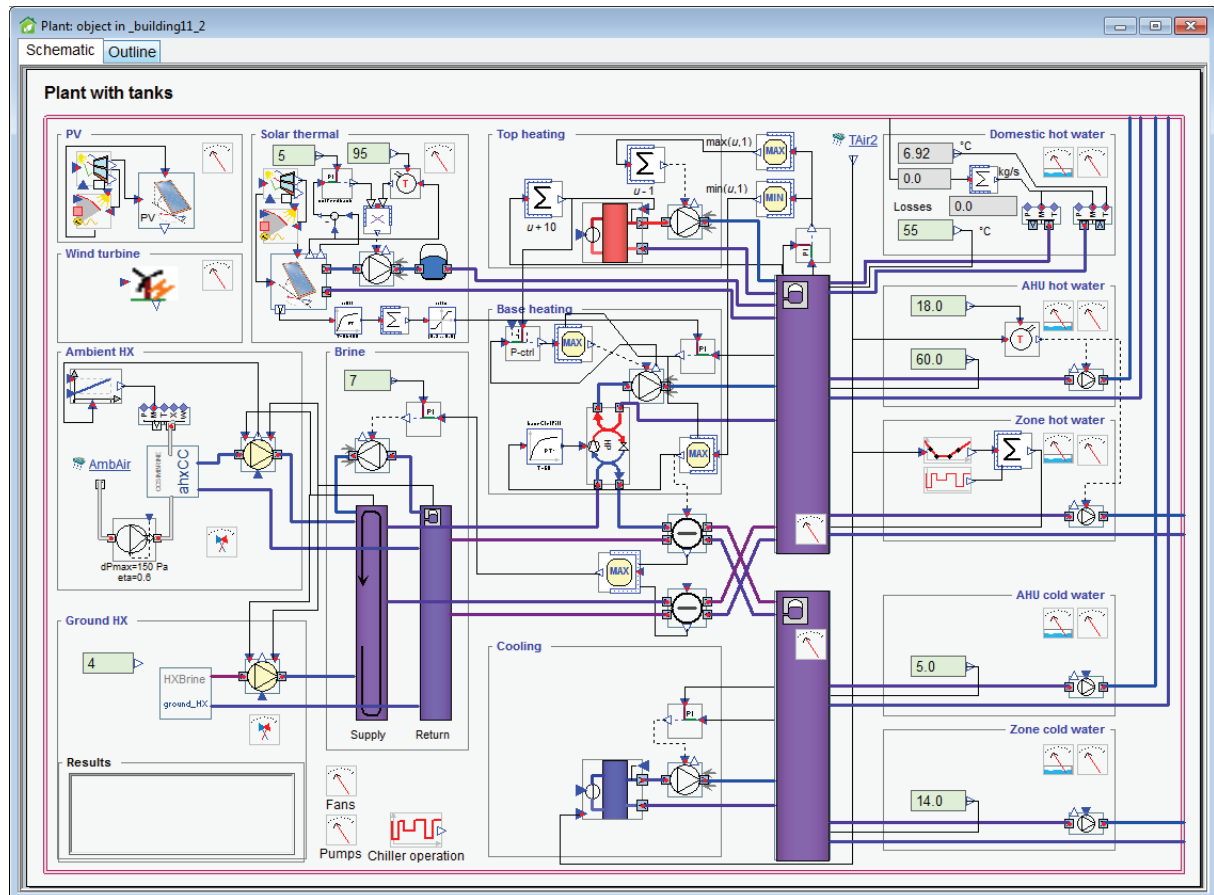


Figure 3.4 The ESBO Plant of Fig. 3.3 viewed at Advanced level of IDA ICE

Organized around the two water tanks, the top one for heat and bottom one for cold storage, are the various circuits that draw or feed tempered water from and to the tanks. Let us start with the client circuits to the right, starting from the top right corner with the DHW circuit.

Domestic hot water is drawn from the hot tank, by the leftmost PMT-object, which has a mass flow signal into it that gives the required DHW mass flow. The make-up water from the water mains is furnished by the rightmost PMT-object and a given temperature (which is computed to be the yearly average temperature of the current climate file). Each client circuit, including this one, feeds a temperature setpoint into the tank. In this case this is the temperature setpoint for the DHW (55°C).

As a simplification, no separate “tank-in-tank” has been defined for the DHW in the default configuration. The heat exchange between DHW and surrounding water is regarded to be infinite and immediate.

The meter objects to the top right are used to record results about DHW production.

Next below is the **AHU hot water** circuit, which is connected via a pump to the air handling unit, providing it with heated water at given pressure and temperature. Similarly as for the DHW, a temperature setpoint (60°C in the example) informs the tank about the temperature that is required by this client. Also in the AHU box is an on-off controller for turning off the hot water circulation to both AHU and zones when the ambient temperature is higher than, here, 18 degrees. Actually a 3°C deadband is applied to this switch (only visible if the component is opened) to avoid frequent switching events.

The **Zone hot water** circuit is similar, but has a more elaborate arrangement for the calculation of the temperature setpoint, allowing it to be a function of ambient temperature as well as a of a night set back schedule.

The **AHU cold water** and **Zone cold water** circuits connected to the cold tank are found in the lower right corner. They have in this example simple fixed setpoints, 5°C and 14°C, respectively, throughout the year.

On the production side, in the leftmost upper corner we have the **PV** (photovoltaics) production circuit. It is, by default, not connected to any other model, but simply receives appropriately shaded sunlight from the components feeding into it and converts it to electricity, the amount of which is recorded by the connected meter.

Similarly, the **Wind turbine**, just below the PV, will generate electrical power only but in response to wind velocity. This power is currently also just measured and reported.

To the right of the PV circuit, the **Solar thermal** collector model is located, with a circulation pump and a separate expansion vessel. This separate brine circuit feeds into a heat exchanger that is located at the bottom of the hot tank. In-tank heat exchangers are also served by idealized heat lance technology. In the default set-up, the solar collector will never prioritize the production of high-temperature DHW, but a fixed (in the example 5°C) temperature difference is upheld in the circuit by a PI controller connected to the pump, instead maximizing the collected amount of heat. When there is no sun, the pump will not operate because the temperature difference over the collector will be negative or below 5°C even at the minimal circuit flow which is always maintained.

A separate monitoring control circuit at the bottom of the Solar thermal area keeps track of collected heat over a longer period of time (days). This signal is used to determine how much the heat pump should be utilized. During the winter, when there is little hope for sun, the heat pump will always attempt to keep the tank as full as possible, to avoid using the Top heating.

Next to the right is the **Top heating** circuit, which is the backup for keeping the top of the tank at the maximal required client temperature. It feeds directly into the tank water, and since expansion vessels are included in the tanks, only a pump is required in this circuit. The control circuit measures the top level water temperature and asks the top up heater to heat, provided the Base heating is already fully engaged.

Next below is the **Base heating** circuit, where the condenser circuit of the brine to water heat pump feeds into the tank directly. The heat pump and the condenser circuit pump are controlled by a PI controller (baseCtrl) which attempts to keep the *fill ratio* of the tank at a given setpoint (which is computed by the Solar thermal monitoring circuit.) The *fill ratio* is defined as the degree at which the tank is filled with water at the highest required setpoint, i.e. if all water is heated to the highest setpoint, the fill ratio is 1, while if the whole tank holds the ambient temperature (20°C), the fill ratio is zero.

Furthermore, the base heating heat pumps will have a speed limitation of the condenser pump circuit that will be active when the heat pump should prioritize hot water production, i.e. the pump speed is limited in order to force the condensing temperature to increase.

On the evaporator side, the brine to water heat pump is connected to the **Brine** circuit. The brine circuit is a more unconventional design and we will explain how it works here. The basic idea is that all free heat and cold sources are connected in parallel to a single brine circuit. The circuit will alternate to feed units that require heat (such as the present heat pump) and cold, i.e. if free sources are available at the same time for both heating and cooling and both a heating and a cooling need exists, a choice has to be made as to which one of these will be satisfied. The idea behind the design is that this collision of interests occurs only rarely.

To illustrate the function of the brine circuit, let us look at what happens with the present base heating evaporator brine flow. Once the base heating heat pump starts to operate, the temperature of the return side brine will drop after the condenser. First the brine passes through a “PMT tap” component, which also receives a signal to open the circuit through the evaporator as the heat pump starts. Should it be beneficial to run the cooled water through a heat exchanger at the top of the cool tank, this is done. After returning back through the PMT tap, the brine is returned to the return brine manifold in the Brine box.

In the event that no free supply circuits exist or are able to operate, and the evaporator was able to discharge to the cold tank, the flow in the brine circuit will be upheld by the pump in the brine box, which monitors both PMT tap components for possible beneficial flows. More commonly, one or several of the free supply circuits will instead pick up the heating need of the heat pump evaporator. Let us look at one of these, the ambient heat exchanger.

The current **Ambient HX** circuit, found immediately to the left of the brine box, is in the example a fan assisted ambient heat exchanger, that is able to both cool and heat the brine when needed. All of the free supply circuits rely on a special pump/controller object “FreeSupCtr” that monitors the temperature difference between the supply and return brine manifolds. When the temperature of the return manifold drops, it is a signal that heating is required, and if the free source circuit is able to meet such a need, it will start its flow (and in the example, the fan of the ambient hx). The circuit will, in heating mode, be operated at a degree which keeps the contributed flow above (by default 5°C) the brine return. The operation will be continued until it is no longer beneficial, for example because the fan power is getting close to the useful delivered power.

All Free supply circuits operate independently of each other in trying to satisfy the current predominant need. Any circuit which can help rise the supply side brine temperature will be operated in a heating situation.

The free supply circuits can also be operated without any condenser or evaporator in the loop. Suppose for example that the bottom of the hot tank has a temperature which is well below outside air temperature. In this situation, the PMT tap component will open its circuit and the ambient heat exchange circuit will start charging the tank directly. This type of situation is of course much more common when it comes to cooling; the free sources will then often be able to directly feed into the cold tank.

In the example, the **Cooling** brine to water chiller operates in a similar way. Any useful heat after the condenser is directly tapped into the hot tank, a free supply circuit will pick up the cooling need and start to operate.

There is nothing that formally prevents both the heat pump and the liquid chiller in the example to operate in parallel, both cooling and heating the brine circuit simultaneously. In

this situation, the brine pump will initially keep the flow in the circuit going but after some time one of the two compression cycles will “win” and create a net cooling (or heating) need, which then is likely to be fulfilled by a free supply circuit.

The present example, with two brine connected vapor compression machines and multiple free supply circuits represent the most complex type of system that can be described. In all other situations, some of these components or connections are missing, creating a simpler system schemata with a lower number of operation modes.

3.2. *The air handling system*

In the default configuration, the air handling system consists of the following components (Figure 3.5): supply air temperature setpoint controller (1), exhaust fan (2), heat exchanger (3), heating coil (4), cooling coil (5), supply fan (6), schedule (7) for operation of both fans and a schedule for the operation of the heat exchanger (8). The unit provides temperature-controlled air at a given pressure. Some key parameters of individual components are presented in the form; open them to edit.

The supply air temperature setpoint is connected to the heat exchanger and to both coils. All three components have separate ideal control circuits, which independently strive to maintain the setpoint. After the coils, the supply fan raises the supply air temperature further by either a fixed number of degrees (default) or by depositing motor and drive losses to the air stream.

In the setpoint controller, three methods are provided for setpoint selection. In the default AHU, the setpoint is set to constant, 16 °C. The second alternative is to let the setpoint vary with time according to a schedule. Thirdly, an option is available where the setpoint is calculated as a (user-defined) function of outdoor air temperature. Note that if for example the chiller (that supplies the cooling coil from the plant) has been turned off or has insufficient capacity, the supply air will not be cooled to the setpoint.

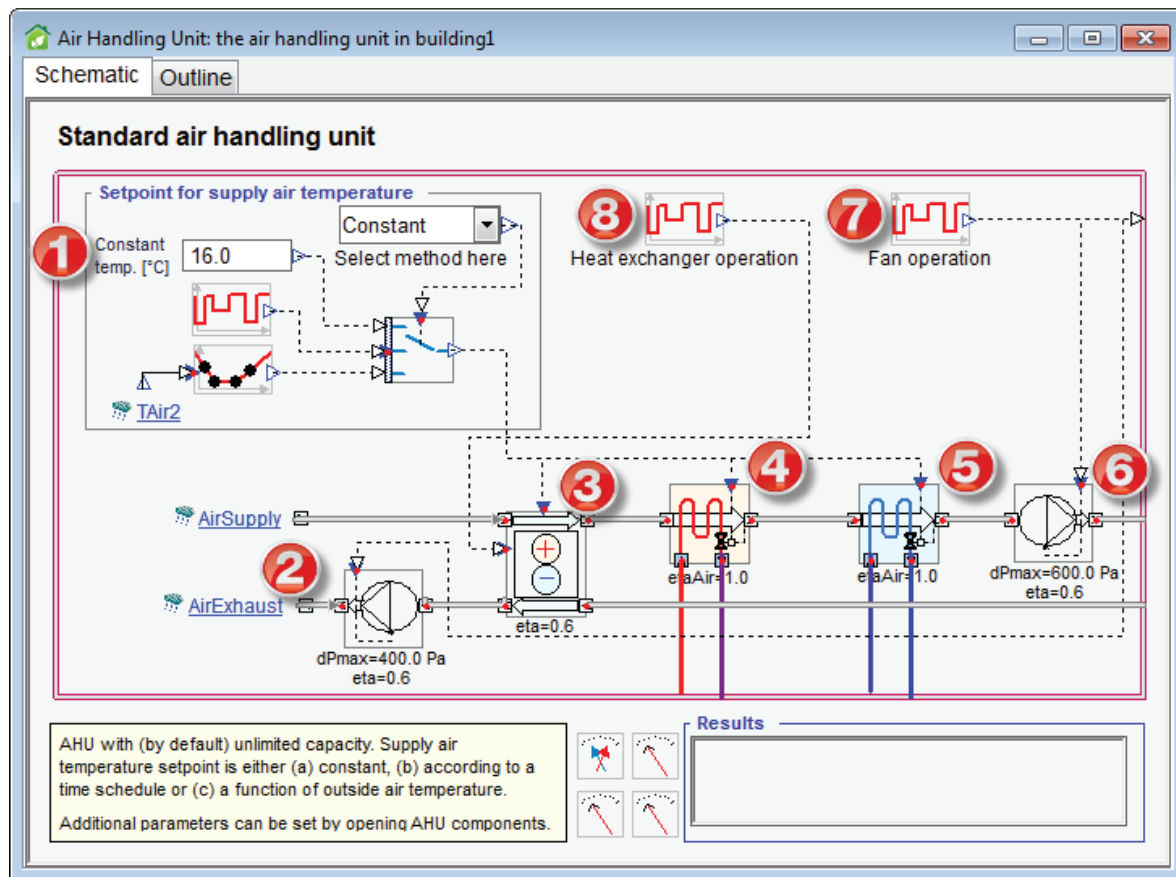


Figure 3.5 Default air handling unit

The heating coil (4) has two important parameters: the air side temperature effectiveness and the desired water side temperature reduction. Capacity control is achieved by adapting the actual effectiveness up to the given maximum level. The necessary water flow is calculated and the water temperature is reduced, if possible, by the desired number of degrees. There is no bypass on the liquid side; control is achieved by restricting the water flow.

In the default configuration, the temperature effectiveness is set at 1.0. There are basically two situations when it may be desirable to change this to a more realistic value: when sizing the actual coil by means of simulation experiments, or when making energy calculations in cases where the heat generation efficiency is dependent on temperature conditions (nearly always for the ESBO Plant.) In addition, the simplest and quickest way of removing the entire coil is to set the effectiveness to zero.

The cooling coil (5) works in the same way as the heating coil, but is mathematically more complicated because air dehumidification is calculated. For wet operation the given effectiveness is defined as $(1 - \text{bypass factor})$, according to ASHRAE's nomenclature. Physically, this means that the state of the cooled air in the psychrometric chart lies somewhere on a straight line between the state for the incoming air and the apparatus dew point temperature on the saturation curve. In the model, the average temperature of the liquid side defines the apparatus dew point. On this line, the given efficiency indicates the status: 0 is no cooling whatsoever, and 1 indicates maximum cooling, which also means that the air at most is chilled to (the arithmetic) mean value of the liquid incoming and outgoing temperatures.

Similarly, the air-to-air heat exchanger (“3” in Figure 3.5) is controlled by adapting the actual effectiveness selected by the model, up to the maximum limit set by the user, so that the setpoint for the supply air temperature is reached (if enough heat is available). The temperature of the (often chilled) exhaust air, which may not fall below a certain level (TEXHOUTMIN parameter), sets another limit. This is to avoid freezing. Note that for rotating heat exchangers, it is usually possible and desirable to cool the exhaust air below freezing. A Capacity parameter for the rated air flow may optionally be provided in the heat exchanger. If this parameter is provided, the actual effectiveness will be adjusted to compensate for lower air flows as well as for unbalanced flow on the supply and exhaust sides.

The heat-exchanger will utilize the return air stream both to heat and to cool the supply air.

The heat exchanger takes into consideration condensation on both the supply and exhaust sides. During wet processes, the given efficiency is interpreted as $(1 - \text{“bypass factor”})$, in the same way as for the cooling coil, but the apparatus dew point for the heat exchanger is defined as the incoming temperature for the opposite medium.

The fans have ideal pressure control with given setpoints and by default constant efficiencies, i.e. they supply a fixed pressure head. In most cases, both these parameters only have significance for calculating fan electricity consumption. By default, the user gives directly the increase in air temperature by the fan (and the system). As an option, the temperature rise may be computed automatically with a given percentage of motor and drive losses being deposited into the air.

For modeling of CAV systems, the fan pressure rise and efficiency at the intended operating point should be entered (pressure can also be given in terms of specific fan power, SFP). For VAV systems, on the other hand, the performance should be adapted for flows below the design point. ASHRAE Standard 90.1 prescribes a way to do this that has been implemented in the fan model. This part load efficiency reduction is activated selecting something else than <unlimited> in the drop box. A rated flow must also be provided for this option.

The fans operational schedule (7) is connected to both the supply and exhaust fans. When the control signal is zero, the fans supply a very low pressure head (for numerical reasons, greater than zero). The fan schedule is also connected to all the air terminals in the zones. When the fans switch off, all terminals are closed. This is to avoid a spontaneous flow through the system caused by the chimney effect.

The fan schedule, like other schedules, gives normally values of 0 (off) and 1 (on) as output signals. However, it is meaningful for the fan schedule to sometimes give other values, thereby forcing ventilation flow. For example, if a value of 1.2 is given, all terminals will supply a 20% higher flow than that selected locally in the zone. This applies to both CAV and VAV systems. In the same way, a value of 0.5 results in half the flow.

The local terminal connection to the central schedule has no counterpart in actual systems, but has been introduced partly to avoid unintentional spontaneous flow in the system, and partly to permit forcing the flow in case of CAV.

The heat exchanger is similarly controlled by a schedule (8). To turn it off, the schedule should give the value 0. A value 1 turns on the heat exchanger. Any value between 0 and 1 will be interpreted as using a corresponding fraction of the heat exchange surface.

3.3. The zone models

For the HVAC systems, the presentations for the standard and advanced user interface levels are the same (for the distinction between standard and advanced levels, see Section 5). However, in the case of the zone models, the presentations are completely different. This section explains the principal features of the zone models, and also provides a brief overview of the advanced level in this context. Actually, most users will never have to deal with the advanced level, but it is useful to know it exists, and the physics are more easily explained from the point of view of the advanced level.

IDA ICE provides two different zone models. One of these, the climate model, is quite detailed - it may for example calculate a vertical temperature gradient. The second model, the energy model, has a more conventional level of precision, and is based on a mean radiant temperature. Both zone models are based on the same description of the building, given in the standard level. All models of components in and around the zone, such as windows, radiators, controllers, leaks, terminals etc., are common to both the energy and climate models. The climate model is currently available only for zones with a rectangular geometry. From version 4, the energy model is default for new zones, but this can easily be changed in the Defaults form. More details about the mathematical models can be found in *Models for Building Indoor Climate and Energy Simulation*¹ or by studying the NMF code in the Code tab of the component window at the advanced level.

Figure 3.6 shows a schematic view (advanced level) of a zone with the climate model. To access this window, press Build model in the Simulation tab, after entering all the information required in the standard level. In this case this includes selection of climate model, an ideal heater, a cooling panel and external shading. After the advanced level model has been generated, the user is able to select between General (standard level) and Schematic (advanced level) tabs for zones as well as for the building. The various component groups are numbered in the figure as follows:

1. Supply and exhaust air terminals
2. Ceiling/Floor²
3. Air leak to ambient
4. Solar irradiation and external film coefficient, external wall
5. One external and three internal walls
6. Window and shading calculation components
7. Proportional controller for occupant automatic clothing adaption
8. PI controller for ideal heater
9. Cooling panel with controller and ceiling section behind
10. The actual zone model in which radiation, convection and loads etc. are modeled
11. Post processing components for results capture.

¹ which can be downloaded from the User's page: Help menu, IDA on the Web, ICE User Support

² This object is both ceiling and floor, since the model was generated to represent zones above and below with identical conditions as the current zone.

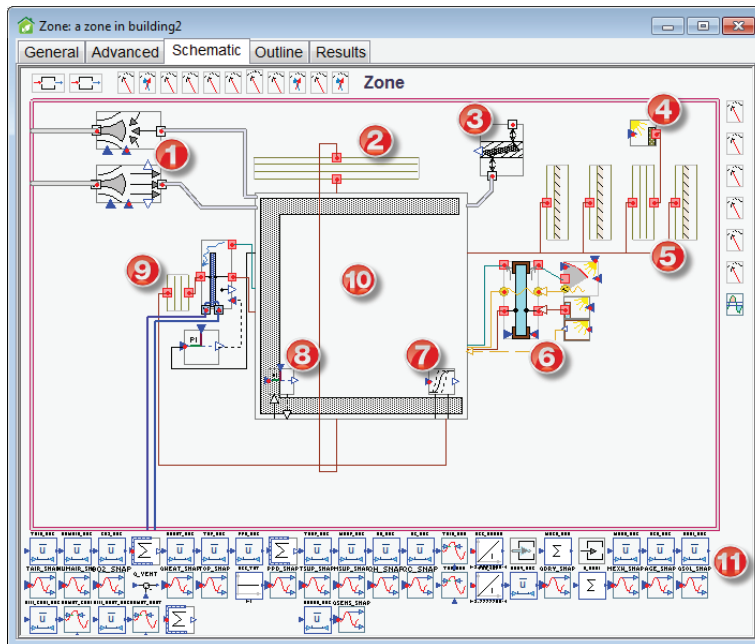


Figure 3.6 Schematic view (advanced level) of Climate zone model

3.4. Solar radiation modeling

A key issue in building simulation is the treatment of direct and diffuse solar radiation. Let us follow the main steps in the treatment of sunlight entering the building. If synthetic weather is used, ICE first computes direct and diffuse sunlight intensities based on the clearness number given in the Location object. At the advanced level, the components used for this computation are found in the Schematic tab of the building form under the heading Climate processor.

In the next model in the chain, the solar position in the sky is computed and all the signals computed so far are sent to a set of Face models, where the climatic conditions outside each main building surface (Façade) are computed. In this model, the distribution of diffuse radiation in the sky is also computed, by default using the Perez model.

In the next step, solar radiation on an individual object such as a window is computed. Connected to each window model (See Figure 3.6) is a shading calculation model (Shade), that computes the shading of both direct and diffuse light on the receiving surface. This model puts all shading surfaces in one 'bin', including building self shading, shading by neighboring buildings and shading by (possibly movable) objects directly outside of the window (External shading). Diffuse light from the sky is also shaded, but no reflections other than ground reflection are accounted for. Ground reflection for each face can be set from the Property page if the face name is selected in the Floor plan tab. Diffuse radiation from the ground is not shaded by external objects. All external shades are considered to be opaque.

The shade model is very difficult to interact with directly at the advanced level, since each surface has been subjected to several coordinate transformations. The actual shading factors are precomputed for all (plausible) solar locations and are stored as parameters in the shade model for the simulation.

Once in the window model, diffuse and direct light are reflected and transmitted depending on the window model used. The standard window model uses a fixed curve for the angle

dependence. Integrated window shading (internal, interpane or external shades in the plane of the window) will reduce radiation by multiplying the basic window parameters. It may also convert direct light to diffuse.

The detailed window model, makes a layer by layer computation of multiple reflections and each layer temperature is computed. For glazings where all the glass panes and the integrated shading have spectral data, the optical calculation in the solar range is made for each wavelength and the values are then integrated to average values according to EN 410.

Once inside the zone, diffuse light is spread diffusely, while the exact target location of the direct light beam is computed. However, the whole surface of the window is considered as the light source, not just the portion of the glass which is actually not shaded by external objects. After the first reflection on a zone surface, the direct beam is spread diffusely in the room. And also here, the whole surface that is hit is regarded to reflect with equal intensity, not just the lit portion of this surface.

Internal windows and open doors transmit light in a similar way as an external window, i.e. the whole opening is regarded as being lit, even if only a small part of the door receives direct sunlight. The light intensity is of course adjusted accordingly. Similarly, light that enters through one window and exits through another external window, for example in a corner room, is treated in a physically reasonable way.

3.5. Airflows

This section covers the models and input which determine the airflows through the building. *IDA ICE* enables the user to take account of natural ventilation, i.e. flows driven by wind pressure and the stack (chimney) effect.

In the simplest case, each zone has three paths for airflows: through the supply and exhaust terminals and via leakage through the envelope. When two zones are placed adjacent to each other and there is an opening in the common wall between them or when windows are open or when additional leaks have been added, additional flow paths are created. However, let us start with the simplest case.

The mechanical ventilation terminals are always of VAV (Variable Air Volume) type. This means that as long as there is sufficient pressure head from the fans, a given flow is maintained as requested by the zone itself. In CAV (Constant Air Volume), the control signal is kept constant and a constant flow will be maintained, irrespective of pressure. In a VAV system, a controller regulates the flow with respect to temperature, carbon dioxide, humidity or pressure levels in the zone.

Since, in the simplest case, flows through two of the three paths are given, the size of the third flow, through the leak, is important only for the pressure in the zone, and not for net flows. Note that if the size of the leak is much too small, an unrealistic pressure may build up in the zone. Such a pressure may become so large as to affect the air psychrometric calculation routines and may then be reported as a condensation problem, when, in fact, it is a pressure problem.

It is also possible to define additional given in/exfiltration flows, i.e. balanced flows into and out of a zone. Since these incoming and outgoing (mass) flows are always equal, they will not

affect the pressure of a zone. The given in/exfiltration will only act to exchange heat, moisture and CO₂ with the ambient.

3.5.1. Air flow input forms

Figure 3.7 shows the zone form. The input fields in the Ventilation section govern air flows. Also involved are Controller setpoints. Clicking on the link field opens the dialog (see Figure 3.8).

Name	Type	Wetted area, m2	Connect to	Azimuth, Deg	Slope, Deg	Construction	U-value, W/(m2 K)	Thickness, m	Layer material	Layer thickness, m	Layer material	Layer thickness, m	Layer material	Layer thickness, m	Layer material	Layer thickness, m	Layer material	Layer thickness, m
Floor	Int. floor	10.0	None	0.0	0.0	Defa...	2.385	0.175	Floor...	0.005	L/W c...	0.02	Concr...	0.15				
Ceiling	Int. cei...	10.0	None	180.0	0.0	Defa...	2.385	0.175	Concr...	0.15	L/W c...	0.02	Floor...	0.005				
Wall 1	Int. wall	6.5	None	0.0	90.0	Defa...	0.6187	0.146	Gyps...	0.026	© Air i...	0.032	Light i...	0.03	© Air i...	0.032	Gyps...	0.026
Wall 2	Int. wall	10.4	None	90.0	90.0	Defa...	0.6187	0.146	Gyps...	0.026	© Air i...	0.032	Light i...	0.03	© Air i...	0.032	Gyps...	0.026
Wall 3	Ext. ...	5.0	Buildi...	180.0	90.0	Defa...	0.5372	0.27	Render	0.01	L/W c...	0.25	Render	0.01				
Wall 4	Int. wall	10.4	None	270.0	90.0	Defa...	0.6187	0.146	Gyps...	0.026	© Air i...	0.032	Light i...	0.03	© Air i...	0.032	Gyps...	0.026

Figure 3.7 The zone form.

The control action of heating and cooling depends on the controller used in the actual device. Defaults are P control for radiators and PI for most other room units.

* when both VAV and other means of cooling have been defined, VAV is used first and setpoints of other room units are offset by 2.0 °C. (Change globally in System Parameters)

Figure 3.8 Control setpoints

In the zone form, the user can select System type (CAV, VAV with temperature control, VAV with CO₂ control or VAV with humidity control, VAV with both temperature and CO₂

control, and VAV with pressure control, schedule controlled VAV etc.). For CAV, the required airflows are given directly in the zone form³. Leak area gives the size of the combined envelope leak⁴. (In the model, the combined envelope leak area is distributed on all external walls, 1 m above floor level. The height is only relevant for natural ventilation situations.) Both Leak area and Given additional in/exfiltration are by default computed automatically based on information for the whole building that is given in the Infiltration form which is reached from the Building form.

In the combo box Controller setpoints (Figure 3.7), the user can select an object with a collection of relevant zone-level control setpoints. The Control setpoints dialog (Figure 3.8) provides input data for zone climate quality requirements. Here, only those parameters, which are of importance for airflows, are discussed. For a VAV system, the given minimum value of Mechanical exhaust airflow provides the lowest allowable airflow, with the maximum value providing the highest one. (In Figure 3.7, an exhaust only VAV system has been specified.) In the case of CAV, as was already pointed out, the desired flow is given directly in the zone form. If the chosen CAV flow falls out of the quality range provided in Control setpoints, a warning is issued when the simulation is started.

The other values in Figure 3.8 may impact on the corresponding VAV control scheme. If the user selects CO₂ control, (see Figure 3.7), the airflow is varied in proportion to the CO₂ content of the air in the zone. For the VAV with CO₂ control scheme, a CO₂ value, which equals or exceeds the given max-value (in Figure 3.8), results in the maximum flow through the exhaust terminal, with a minimum CO₂ value producing the given minimum airflow. The humidity control function is entirely analogous with respect to relative humidity (assuming supply air will dry the zone).

The option 'VAV with temperature control' functions somewhat differently. Here, the maximum comfort temperature value is used. Forcing of the VAV flow begins somewhat below (normally 1 °C) the indicated maximum value (P control). Full exhaust flow (see Figure 3.8) is reached at somewhat above (normally 1 °C) the maximum temperature value. The throttling range is normally 2 °C, but can be selected at the building level under System parameters.

This scheme will assume that the supply air is able to cool the zone, i.e. if there is a need for heat and the supply air is warmer than the zone, this will not be recognized by the controller.

The option 'VAV, temp+CO₂', on the other hand, will be "smart" enough to both heat and cool with the supply air. It relies on PI controllers instead of P ditto, and will therefore not have any offset error. In addition, it will also force air flow if needed to maintain CO₂ level at the maximum limit. (The minimum limit is not used.)

Pressure controlled VAV is normally used for return air flow control where some other VAV method is used to supply air into adjacent zones. It will attempt to maintain zone pressure, within the given range, with respect to ambient pressure using a proportional controller

³ The link "l/s m²" next to the input field enables input in various units.

⁴ Equivalent Leakage Area (ELA) at 4 Pa and Cd=1. ASHRAE Fundamentals provides a number of values regarding leaks according to this definition, as well as conversion methods between various representations of leak sizes.

(measured as pressure drop in the local ambient leak). In the example in Figure 3.8, the zone is maintained between 10 and 20 Pa below ambient pressure.

3.6. General information regarding zone cooling and heating room units

Local heating or cooling is supplied to the zone by room units. All room units are listed in the zone form. Some units such as ideal heaters and coolers, do not have a given location in the room. These can be introduced directly into the list in the zone form. Most hydronic units may, on the other hand, be located on a specific zone surface and they are instead inserted by dragging them onto a surface. From version 4.5, most hydronic units can also exist without a specific position in the zone, i.e. they can be dragged directly into the zone form. Note, however, that all radiative units still require an explicit surface area. (Given on the Properties page, when the input form of the unit is active.)

The temperature setpoint for cooling devices is normally taken directly from the Control setpoints, max value for Temperature⁵ (see Figure 3.8). The corresponding value for heaters is the Temperature min value. However, from version 4.5, it is also possible for the Expert edition user to define any controller for an individual device.

3.7. Ideal heaters and coolers

Ideal room units should be used to condition the zone when no detailed information about an actual room unit, such as a fan coil or active chilled beam, is available or this amount of detail is unmotivated. They have no given physical location on any room surface *and are not connected to the plant of the building*. They do have a maximum capacity parameter, enabling the user to experiment with limited heating/cooling capacity. However, this parameter should normally be set to a large enough value to always cover any foreseen need. However, do *not* set the value to a totally unrealistic number, e.g., 100 times the reasonable heating load. This will result in poor control action.

3.8. Simple fan coil

From version 4.7, a new room unit, Simple fan coil, has been introduced. It is always connected to the cold water circuit and, optionally, also connected to the hot water circuit.

Fan electricity use is specified as a percentage of cooling (or heating) power. This percentage can be set to zero without affecting the operation of the device. Setting the fan electricity to zero in a Simple fan coil is actually the computationally most economical way of heating or cooling the room by water from the Plant.

The Simple fan coil on the advanced level is actually implemented by using Ideal heaters and coolers in the zone and then simply removing the corresponding amount of heat from the water stream.

⁵ An exception is made when the air system is temperature controlled VAV. Cooling units in this case have their setpoints displaced by usually 2 °C. (Can be given centrally in System parameters). This means that the airflow is first forced to max value, after which the other cooling unit sets to work (assuming that free outside air cooling is often available).

3.9. Hydronic heating devices

Most other hydronic devices use a mathematical model that allows them to have a given position on a zone surface. This enables the computation of correct local radiation near the device (when the detailed zone model is used).

Heat emission from hydronic heating devices is calculated using

$$P = K * l * dT^N,$$

where l is device length and dT is the temperature difference between the water and the zone air. K and N are constants characterizing a device of a certain height (or width for ceiling devices).

Figure 3.9 has a radiator inserted on a wall surface. Its main form has been opened and a dialog for the Device type has been opened from the form. Often the values of K , N and $Height$ come from a database, which in the form can be opened under “Use manufacturer’s data”. The data for the Device type can also be altered and saved by the user with a new name. The user must then only enter the surface area (given graphically) and the Design conditions, from which the mass flow rate is calculated.

The warm radiator surface that is exposed to the zone is defined by the box that is drawn when the unit is inserted on a wall surface. The size of this box has meaning for the heat transfer. While the total emitted heat is always given by the expression for P , the division between radiation and convection is calculated based on the surface temperature - which is connected to the water temperature - and the given exposed surface. The zone “sees” a warm surface and calculates the radiation and convection from this surface. The main part of the remaining heat (needed to complete P) is emitted convectively directly to the air (corresponding to convection behind a radiator). A small part of the heat goes to increase the temperature on the portion of wall behind the heating device. The heat transfer coefficient between the device surface and the wall behind is considered in the basic case to be completely dominated by radiation and is calculated by the model.

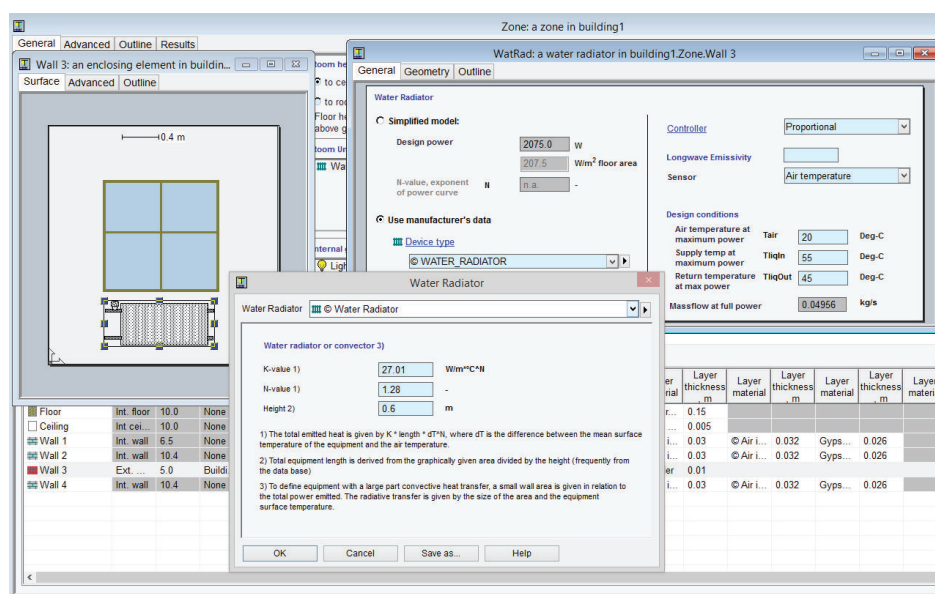


Figure 3.9 A radiator on an external wall, its standard form, and a dialog for the Device type.

To facilitate adding new heating devices without direct knowledge of K and N , a Simplified model data input is given (see the form in Figure 3.9). The values for design mass flow and K are calculated from the power given by the user at the *specified* Design conditions. The user also provides a value for N in this case. Note that K is calculated from the information given in the alternative input. This is thereafter the value for K . If the size of the graphical box is later changed, the device is likely to have an unintended maximum power. Note also that the actual maximum heating capacity of the device will vary with actual room and supply water temperatures. The Design conditions are only used to calculate K and have no impact on the simulated supply and return water temperatures⁶.

3.10. Cooling units

The Cooling device is used for radiative and convective units. Cooling units operate completely analogous to waterborne radiators, with the exception of a few differences in input, which are discussed here.

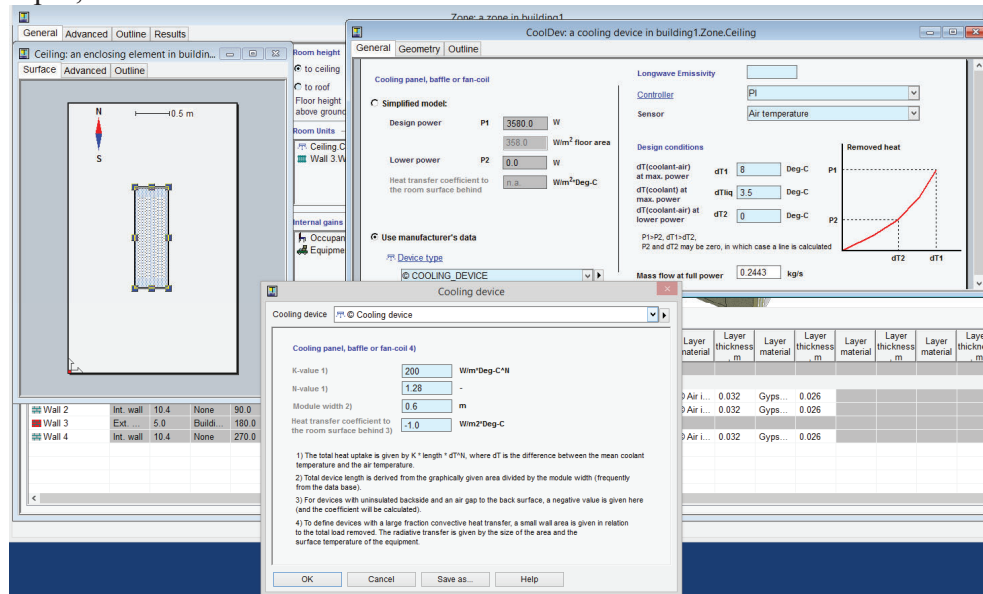


Figure 3.10 A cooling panel on the ceiling surface, its standard form, and a dialog for the Device type.

Here, the height of a radiator corresponds instead to a *Module width*, to which K and N refer. The total length is calculated as the given box area divided with *Module width*⁷. There is a difference in that the heat transfer coefficient between the back of the device, and the surface behind (often the ceiling) is given directly in the main form. If an (arbitrary) negative figure is entered, the heat transfer coefficient is calculated in the same way as for the heating device, i.e. as if all heat transfer was done by radiation. This is a good approximation for a device that has no insulation at all.

⁶ The actual water supply temperature is given by the boiler; the return temperature will vary with the actual massflow through the unit, the current room temperature and the size (K value) of the heating unit. However, if actual temperature conditions are made to coincide with those given in the dialog, exactly the given power will be emitted.

⁷ The same is done for the radiator.

The dialog for alternative input has somewhat different parameters for cooling units. Absorbed power and temperature differences between air and water may be given for two points on the power curve. For max power, the temperature rise of the water is also given.

3.10.1. Active beams

Active beams serve both as supply air terminals and as cooling devices with significant convection. Their performance depends on the amount of supply air that is passed through but they normally retain a heat transfer contact with room air also in the case of zero supply air flow. The radiative coupling with the room is neglected in the present model.

Two input data options are available: Simplified and Manufacturer's. The latter means that the performance parameters K and N are given as functions of air flow. This alternative is mostly used when data is automatically imported from an on-line manufacturer's database. The Simplified option is based on two user supplied performance points, at design conditions and at zero flow (Figure 3.11). N is for this case set to 1.5.

Beam: a device in building2.Zone

General Outline

Simplified input data to chilled/heated beam

☐ Use manufacturer's data
☒ Simplified model:

When using the detailed model, the power at the given condition is calculated from the manufacturer's data and shown in this form.
 To edit manufacturer's data, switch to the outline tab.
 In the simplified model, the power parameters are user-supplied. No manufacturer-supplied data are used.

	Cooling	Heating	
Power at design air flow	0.0	0.0	W
Power at zero air flow	0.0	0.0	W
Design air flow	0.0		L/s

Design conditions

dT(coolant - zone air) at max power	8.5	20	Deg-C
dT(coolant) at max power	3	10	Deg-C

Controller: PI

Sensor: Air temperature

Figure 3.11 Active beam form

In the constant flow (CAV) case, the given Design air flow is regarded to pass through the beam whenever fans are running. For VAV, the constant Design air flow will pass through the beam whenever there is sufficient air into the room and surplus air will feed directly to the room without first passing the beam. If the VAV flow is insufficient to serve the beam with the full requested Design flow, the flow through the beam will be reduced accordingly.

The total Design air flow through all beams must not exceed the total for the zone. If beam air flow is less than the requested total for the zone, the remaining part is regarded as being supplied through conventional terminals.

When the flow to the room is increased by forcing the central fan (or similarly reduced) the beams will still keep their requested Design flow, as far as possible.

3.10.2. Heating/Cooling floor

If a floor heating/cooling object is inserted on the floor of a zone, the floor construction for this area is divided into two parts, above and below the heated layer. Between the two, a heat exchanger model is inserted corresponding to the piping layer. Quite often a floor heating circuit will heat the room below almost as much as the room it belongs to.

The floor coil model assumes that the active layer can be treated as an infinitely conductive plane in the floor slab, i.e. all 2D effects are disregarded. Heat transfer is calculated with a logarithmic temperature difference between the fluid and this plane of constant temperature. The user supplied total heat transfer coefficient between the fluid and the plane, includes

1. Convection between medium and tube wall
2. Heat conduction through the tube walls
3. “Fin efficiency” corresponding to the distance between immersed tubes or actual fins.

The modeling approach will in steady state correspond to the Resistance method of the standard EN 15377-1.

The floor coil circuit can have its own three-way valve and pump circuit keeping the massflow constant (default). Emitted power is then controlled by varying the supply water temperature. PI, P or on-off control can be selected (PI is default). A further alternative Always on is also available, which will keep both the boiler mass flow and the coil circulation massflow permanently at their design values. Control must then be maintained by the boiler temperature controller.

In the case where no separate coil pump circuit is used, the four control options will instead act by limiting the massflow through the coil, or keeping it constant at design conditions in the Always on case.

hc-floor: object in building1.Zone.Floor

General Geometry

Floor heating and cooling (temperature control)

	Cooling	Heating	
Design power	40.0	40.0	W/m²
	20.0	20.0	°C
dT(water) at design power	5.0	5.0	°C
Controller	PI		
Sensor	Air temperature		

The design power and temperature differences are used to calculate supply massflows. The actual emitted power may become smaller if the heat resistance in the floor construction is too large.

Coil massflow

☐ Flow control (2 way valve)

☒ Temperature control (3 way valve). Flow given by Design power and this DeltaT: 3 °C

Location in slab

Depth under surface: 0.02 m

The floor heat coil becomes a tempered layer at the given depth in the floor construction, which is defined elsewhere for the whole floor.

Heat transfer coefficient

H-water-pipe-fin*: 10 W/(m²·K)

* In the nomenclature of the EN 15377-1 standard:
 $1/H\text{-water-pipe-fin} = R_w + R_r + R_x + R_t - R_z$

Normally set to 6 for aluminium fins in a wooden construction and to 30 for pipes immersed in concrete. However, the total heat transfer is normally largely determined by the resistance in the floor construction above and below the coil, in which case this parameter becomes less important.

Figure 3.12 Floor heating form

4. CAD and image import

In *IDA ICE* it is possible to import CAD objects and image files. These can be used as a base when creating the simulation model, or as shading elements that cast shadows onto the simulation model. There are three categories of CAD objects and image files; building information models (BIM), CAD and vector graphic files, and image files.

BIM files contain 3D geometry as well as properties for rooms, walls, windows and materials etc. An *IDA ICE* model, i.e. building bodies, zones and windows etc., can be automatically created from the geometrical information. Furthermore, the properties of objects in the BIM file can be mapped to the corresponding objects in the simulation model. The 3D geometry of a BIM file can also be selected to shade the simulation model.

CAD and vector graphic files contain 3D or 2D geometry. A section of this geometry is shown as lines in the floor plan tab and these lines can be used to snap building bodies and zones etc. in the floor plan. Building bodies and zones can be automatically created from graphic files if the imported geometry consists of volumes enclosed by polygon surfaces. 3D CAD objects can be selected to shade the simulation model.

Image files contain raster (bitmap) images. These are shown in the floor plan tab when the section is close to the location of the image. The images can be used as a background when drawing building bodies and zones in the floor plan or when inserting windows and shading objects in the 3D view.

4.1. Supported file formats

4.1.1. BIM

Industry Foundation Classes (*.ifc)

4.1.2. CAD and vector graphic files

AutoCAD (*.dwg⁸, *.dxf, *.dwf)

SketchUp (*.skp⁹)

3D Studio (*.3ds)

Wavefront (*.obj)

Computer Graphics Metafile (*.cgm)

Corel Presentation Exchange (*.cmx)

MicroStation DGN (*.dgn)

Micrografx DRW (*.drw)

Gerber File Format (*.gbr)

Scalable Vector Graphics (*.svg)

Printer Command Language (*.pcl, *.prn, *.prt)

⁸ *IDA ICE* supports DWG file formats up to AutoCAD 2004. DWG files of unsupported formats can be converted with the free tool Autodesk DWG TrueView. DWG files are assumed to be two-dimensional, i.e. any 3D geometry is flattened to 2D at import.

⁹ *IDA ICE* supports SketchUp files up to version 2013. SketchUp files of later versions can be saved as version 2013 from SketchUp.

Macintosh PICT (*.pct)
HP-GL/HP-GL2 (*.plt)
WordPerfect Graphics (*.wpg, *.vwpg)

4.1.3. Image files

Bitmap (*.bmp)
JPEG Interchange Format (*.jpeg, *.jpg)
Portable Networks Graphics (*.png)
ZSoft PC Paint (*.pcx)
Tagged Image File Format (*.tiff, *.tif)
Adobe Photoshop (*.psd)
Truevision (*.tga)
Windows Meta File (*.emf, *.wmf)

4.2. *Importing IFC files*

In IDA ICE it is possible to import 3D building information models (BIM) via IFC files. Most 3D CAD applications can export architectural data in the IFC format.

The most important information that is transferred is geometrical data, i.e. the shape and position of zones, windows, doors, building faces etc. Zones in IDA ICE are automatically created from so called space objects in the IFC model. It is not sufficient that the CAD model only contains wall objects, spaces that fill the voids between walls must also have been created, a semi-automatic process in most CAD tools.

IDA ICE can also utilize other types of information in the CAD model, such as space types, window types, wall constructions, should they be present. One can find more detailed information about the ICE IFC implementation on the user's web page: Help menu, Support, Download & info center. Select Documentation on the page and open the document 'IFC Import'.

4.3. *Mapping data from IFC*

Start with a building without zones and select the Floorplan tab. Press IFC... > Import... to select an IFC file for loading. There are some sample IFC files in the installation, normally located in C:\Program Files\IDA\samples\ICE\IFC.

The first task is to map named data objects in the IFC model (if any are present) to corresponding IDA resources. Press IFC... > Mapping... in the Floorplan tab to open the Mapping dialog (Figure 4.1). If wall constructions have *not* been described in detail in the CAD model, select directly Constructions in the Category combo box. This will present a list of all wall types that have been found in the IFC model. Since IDA ICE needs more detailed information about a wall, IFC wall types need to be manually associated with IDA ICE wall constructions. To bind a certain IFC wall type to an IDA ICE construction, select both the IFC wall type and the corresponding ICE resource and press Map to selected. A right pointing arrow in the IFC Data list indicates that the item has been bound. Usually one first has to load relevant IDA ICE resources from the database by pressing Load from Db. To inspect the selected IDA resource, press View.

Repeat the procedure for window types as well. Here, one usually has to first create relevant windows in the ICE database, including internal shadings etc.

If wall constructions have been described in the IFC model with layer thicknesses and material names, one can automatically create corresponding IDA ICE constructions. In this case, one starts instead with binding IFC material names with IDA ICE material resources. Once the materials have been mapped, IDA ICE wall construction resources are created by pressing Import from IFC when the relevant IFC wall type has been selected.

A useful addition, available from version 4.7, is the ability to map IFC space type names to zone templates. This way, if you have created templates for your typical room types, zones will be created with all input data already in place.

Any object in the IFC model which is not explicitly mapped to an IDA resource will be set to its default value, which is given by pressing Defaults on the General tab in the building form.

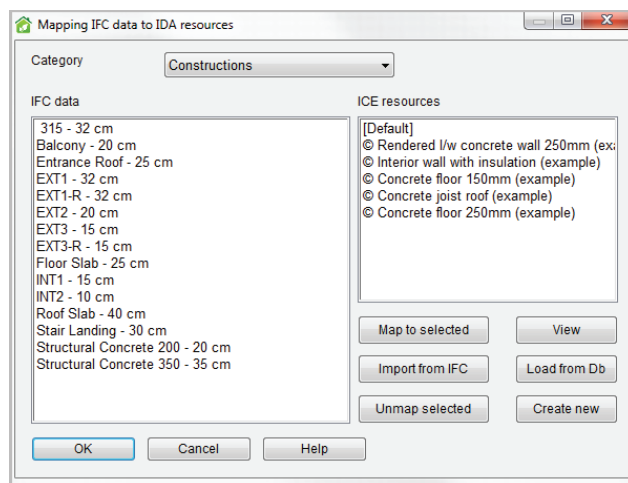


Figure 4.1 IFC Mapping dialog

4.4. Changing position of IFC model

The position of an IFC model can be changed in IDA ICE. Press IFC... > Shift... to open the Shift dialog. In this dialog an offset or desired position can be defined for the IFC model.

4.5. Create zones from IFC spaces

An IFC model may contain more than a single floor. A horizontal section (slice) of the building at a certain level is shown in the Floorplan tab (Figure 4.2). To select a different level, press the button Level: xx m, where xx is the floor height from ground of the current level. In the Level dialog, the building height from ground (Building top) and height coordinate of the floor slab with respect to ground (Building bottom) are also shown, as interpreted from the IFC file. These numbers are not always correct for the user's purpose.

To define which spaces in the IFC model that should constitute a (thermal) zone in the simulation model, click on (select) the neighboring spaces that should be included. (Click again to unselect a space.) Think about zone economy, i.e. do not create more zones than you think is physically motivated for the current study. To create an IDA ICE zone from the

selected IFC spaces, press New zone¹⁰. This will create zones using the currently selected zone template, unless the IFC space type has been mapped to a given template. Try to give as many reasonable defaults as possible for new zones using appropriate zone templates; after creation these values must be edited separately for each created zone.

If you have a good quality IFC-file, you can also - from version 4.7 - instantiate all IFC spaces to zones with a single press of the New zone button without first selecting the space. To do this, select this alternative under the IFC button.

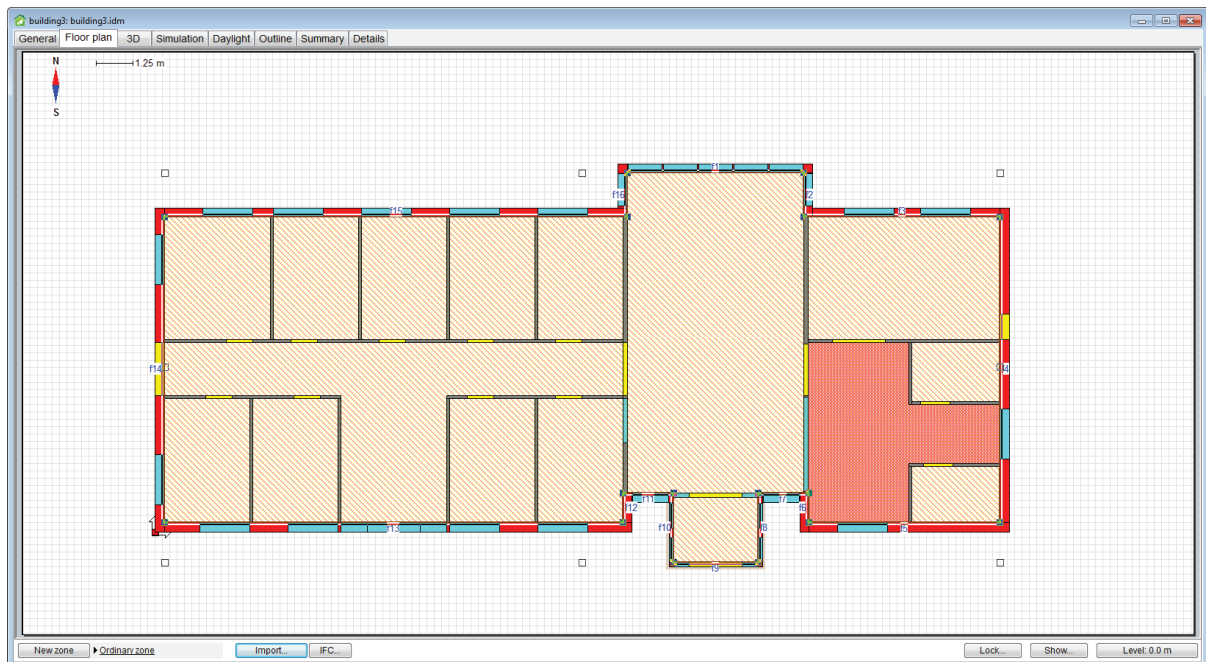


Figure 4.2 IFC model with an ICE zone, a selected IFC space and unselected IFC spaces.

The IDA ICE zones are created from the geometry of the corresponding IFC space(es). If for example a space is taller than the typical floor to floor distance of the building, the corresponding ICE zone will also reach over more than a single floor. One can change the horizontal section level during the zone creation process but it is currently not possible to combine several spaces vertically into a single zone. Note that the Floorplan view displays two models simultaneously, the zones of the created ICE model and the spaces of the IFC model. Both categories of “rooms” can be individually selected and ICE zones can also be opened.

If the IFC model is revised during the ICE modeling project, there is some support for retaining previous work. New IFC models can be loaded while modeling, either replacing the existing model or adding to it, e.g. loading several floors that are in separate IFC files. When a new file is loaded, the user is given the option to replace or add to the current IFC model and to replace or keep mapping information and existing ICE zones.

¹⁰ By default an individual zone is created for each selected IFC space. Optionally, all the selected spaces can be merged (if they have the same floor and ceiling level) into larger zones. This setting is found under the ‘IFC’ button

If the IFC information is incomplete or too complex for some part of the building, the user can simply avoid to instantiate these zones based on IFC background and draw them manually on the floor plan.

4.6. Exporting results to IFC

Results from heating and cooling load simulations in IDA ICE can be exported for merging with IFC models. This is done by choosing “Export to IFC” in the Tools menu after simulation of the heating and cooling loads.

4.7. Importing CAD objects as building bodies or zones

CAD objects can be imported as building bodies or zones if the imported geometry only contains a volume enclosed by polygon surfaces (polyhedron) without holes between the surfaces. The geometry should describe the inner surface of the external walls for a building, and the inner surface of the zone walls for a zone. No other information than the pure geometry of the building body or zone can be included in the CAD object. Click Import... on the floor plan tab and choose Import building body... or Import zone geometry....

Imported building bodies and zones have protected geometry, i.e. their geometry is non-editable. However, an imported building body is fully editable if the imported geometry only has one floor and that floor is horizontal and does not contain any holes, and the geometry does not have any outward leaning walls (surfaces with their exterior normal pointing downwards). This is the same kind of geometry that can be created in the ICE roof editor. Importing geometry as zone will also create a building body of the same shape as the zone.

If a geometry file contains multiple polyhedron geometries, each with a separate color, they are imported as separate building bodies or zones in ICE. If surfaces are placed one wall thickness apart, these are regarded as thermally connected internal walls.

4.8. Importing CAD objects and images as background

CAD objects and image files are either imported with respect to the building coordinate system, and are then moved with the building if the building is repositioned or rotated, or they are imported with respect to the site coordinate system, and remain fixed if the building is repositioned or rotated.

Import a CAD object/image file with respect to the building coordinate system by clicking the Import button on the floor plan tab and choosing CAD and vector graphic. Alternatively, select Import CAD on the Insert menu while the 3D tab is shown.

Import a CAD object/image file with respect to the site coordinate system by clicking the Import site CAD button on the Site object dialog opened by clicking Site shading and orientation on the General tab. Alternatively, select Import CAD to site on the Insert menu while the 3D tab is shown.

To place a CAD object/image at the current mouse pointer in the 3D view, use Right mouse button menu > Import CAD or Right mouse button menu > Import CAD to site.

4.9. Moving and scaling CAD objects and images

A CAD object is automatically scaled and positioned so that it corresponds to the simulation model. The scale and position of a CAD object can be seen and edited by double-clicking on

the object. A section of the CAD object or image is shown in the floor plan tab if the floor plan level is within the bounds of the object. Select this section by clicking on it¹¹. Move and change size of the CAD object/image by dragging/resizing the section.

A CAD object can also be moved in the 3D view. Select the CAD object, hold down the ctrl-key and drag the object. The object moves in the x-y plane. To move a CAD object along the z-axis, hold down the ctrl-key and the shift-key while dragging.

4.10. Shading by imported 3D objects

3D CAD objects can be selected to shade the simulation model. Check the Calculate shadows checkbox in the dialog shown when the object is double-clicked. Every surface of the 3D object that is not fully transparent is included in the shadow calculation. Semi-transparent surfaces are included and shade according to their transparency. Note that it is the surfaces that shade, so for instance for semi-transparent sphere light will pass two surfaces and hence be reduced by the semi-transparency twice.

4.11. Storage of CAD objects

By default the geometrical information of CAD objects is saved in the system file (*.idm). The original CAD file is then not needed after the import. However, if a CAD file is big, very large system files can be created and the performance of IDA ICE can be slowed down. Thus, if a CAD file is big, the option of not saving it in the system file is given in the Preferences dialog, which is opened at import. In that case, only a shortcut to the original CAD file is saved in the system file and the original CAD file needs to be saved in the location specified by the shortcut. If the CAD file is placed in the ICE system folder (the folder with the same name as the system), the shortcut is relative and the CAD file is automatically copied with the system to a new location. Otherwise the shortcut is absolute and the CAD file is not copied with the system. The definition of a big CAD file can be changed in the Preferences dialog from the default of 10000 vertices. The Preferences dialog can also be opened from the Options menu.

¹¹ To select a CAD object, click on the geometry lines. To select an object that is behind another object, press the ctrl-key and click on the object until it is selected.

5. Getting started with the advanced level

The standard level interface covers the most common simulation tasks, but sometimes it is useful to examine other variables than those available here or to replace a component model. To accomplish this one turns to the advanced level interface, where the system is described in a mathematical sense with components containing equations, variables and parameters.

Access to the advanced level is provided for both the Standard and Expert editions of IDA ICE. However, Expert edition users can manually edit (reconnect) component diagrams at the advanced level, while Standard edition users can only inspect, log variables and change parameters. Some of the examples below require the Expert edition, but this will then be mentioned in the introduction.

Work at the advanced level is best presented in terms of demonstration, so the written account here is quite brief. Look also at the User's web page for more information on work at the advanced level, demonstration movies etc.

In some cases the system structures at the standard and advanced levels match each other quite well. The air handling unit, for instance, has different components: fans, coils etc. The same description can then be used for both the standard and advanced levels. The same is true for the primary system. However, for the actual building description entirely different structures are used for the standard and advanced levels.

Most components at the advanced level are described with equations. Components are interconnected by creating equalities between variables that appear on interfaces. An example is the interconnection between the cooling and heating coils in the air handling unit. Both the outflow interface of the heating coil and the inflow interface (terminal) of the cooling coil contain variables for: pressure, temperature, massflow, moisture and carbon dioxide concentrations.

To work on the advanced level, select Build model in the Simulation tab. This creates the schematic view of the current building. If the model has been built previously, Schematic is already available as a view of the system. Figure 5.1 shows the appearance of the air handling unit in the schematic view (which is also the standard view for the air handling system).

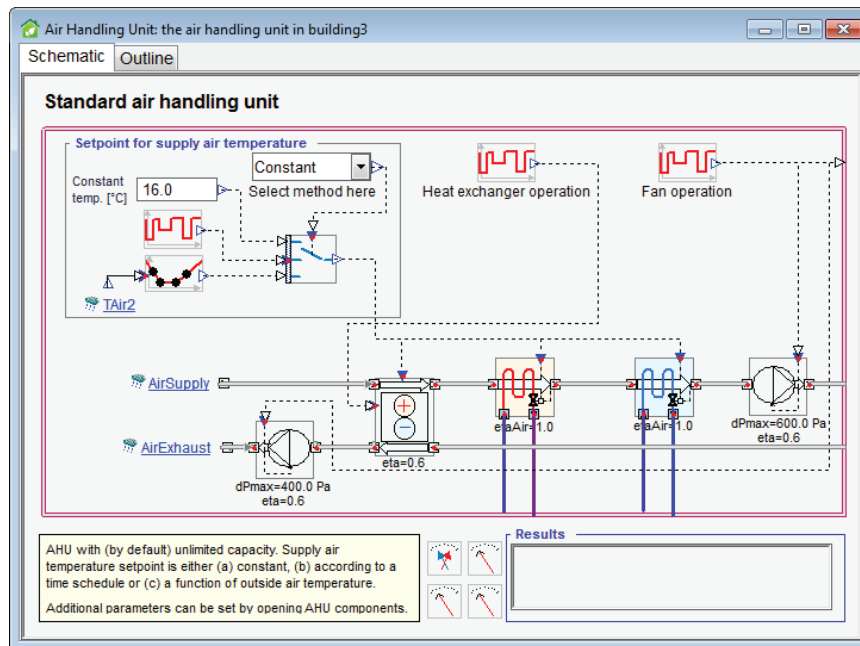


Figure 5.1 The air handling unit.

The behavior of a component is described by equations, variables and parameters. The difference between parameters and variables is that the former will never change their value during a simulation, whereas a variable might. A window area is an example of a "natural" parameter and the room air temperature is always a variable.

The description of most components is done using a special language called NMF (Neutral Model Format). Click on the Code tab of any of the AHU components, to see the NMF code. It contains the following main sections:

Abstract	A brief textual description of the model
Equations	The actual mathematical description (formalized according to a strict syntax but quite readable also for humans)
Links/Interfaces	A description of the ports or terminals of the component. A fan would typically have at least two links for incoming and outgoing air flow and could also have links for power supply and control signals.
Variables	Variables to be calculated by the model.
Parameters	Quantities that characterize the component, e.g. a vector of numbers describing the fan curve.
Parameter processing	Computer code which converts user supplied parameters into those that actually appear in the equations

From version 4, some models in the ICE library are instead described by the Modelica language (www.modelica.org)¹².

¹² The IDA Modelica development environment is not yet publicly released and focus here will be on NMF. The NMF development environment is shipped on request with the Expert edition of IDA ICE.

We will now go through a few examples of useful operations at the advanced level of ICE.

5.1. Example 1: Presenting more data in an existing diagram

Suppose we are interested in the air temperature after the heat exchanger in the air handling unit and would like to see the graph together with the other temperatures of the AHU. Open the AHU window (Figure 5.1) and double click on the connection between the heat exchanger and the heating coil. A small window is opened showing the connection between the two interfaces; SUPOUT (SUPply OUT) of the heat exchanger (hx) is connected to the AIRFLOWIN interface of the heating coil (hc). To see the actual variables of the connection, double click on the box hx.SUPOUT. TSUPOUT is the variable we are looking for. Double click on it to get the form at the lower right of Figure 5.2.

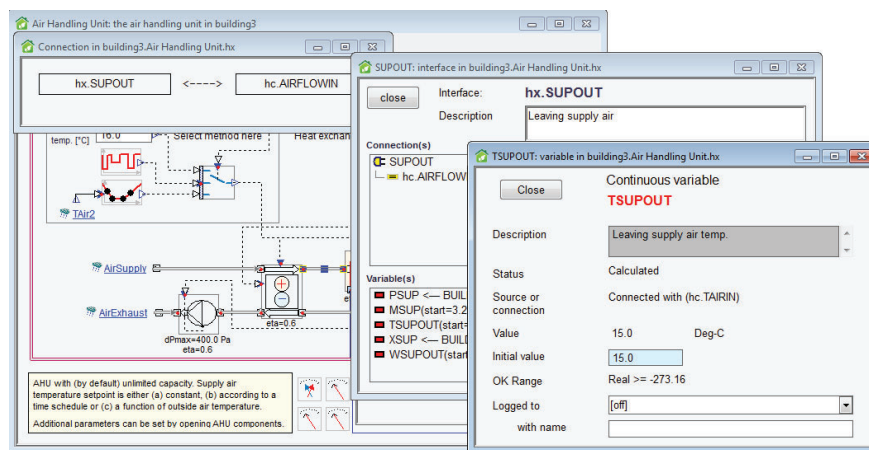


Figure 5.2

All relevant information about the variable is displayed in this window. At the bottom of the window, there is a combo box for logging the variable to a diagram. Select “AHU temperatures” in the combo box and give a meaningful name to this variable in the corresponding field. Make a simulation, and inspect the added graph in the diagram.

5.2. Example 2: Shade control by zone temperature (Expert edition required)

Normally, window integrated shading is controlled by the amount of radiation that penetrates the glazing. However, in some applications it can be useful to let the zone air temperature determine whether shades should be drawn instead. It is possible to define customized controls for shading (and other devices) at the standard level.

We look at a simple case consisting of a single zone with a window in one wall and intend to have an external blind shading the window depending on the zone air temperature and governed by a thermostat. First insert an external blind. Double click on the window to open the window form. Search in database under “Integrated Window Shading” and select “External blind” as Device as shown in Figure 5.3.

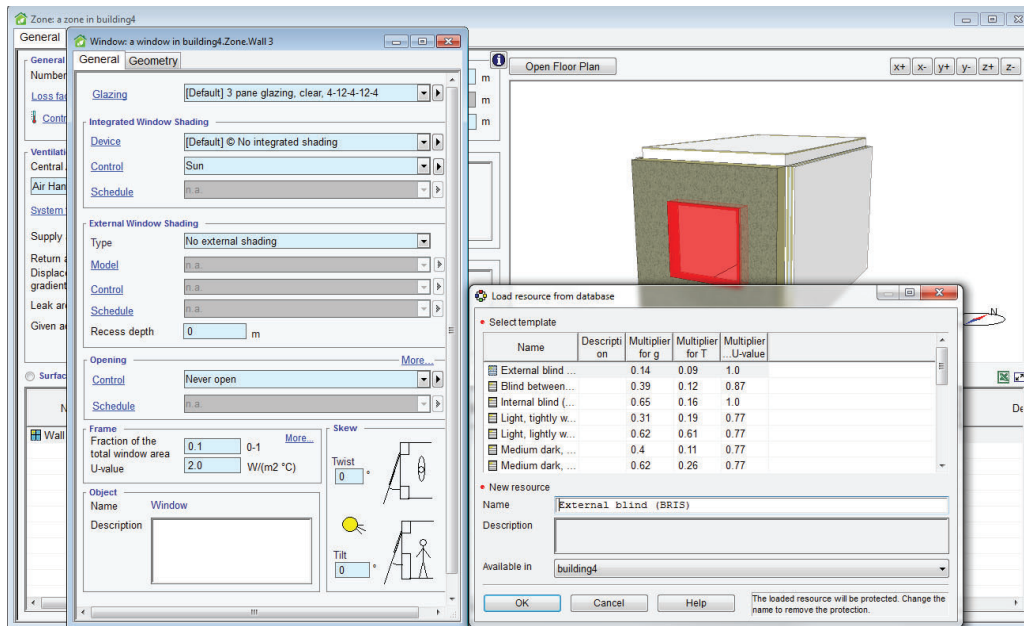


Figure 5.3

Next we define the control system to use. Right below the previous choice in the window form select “New custom control” as Control and give an appropriate name. An empty macro form appears. Drag a thermostat from the Control palette and connect the AirTemp link on the “Zone” box to the measure link on the thermostat as shown in Figure 5.4. Click and hold to draw the connection. Similarly, connect the out signal link of the Thermostat to the macro output “Shading signal”.

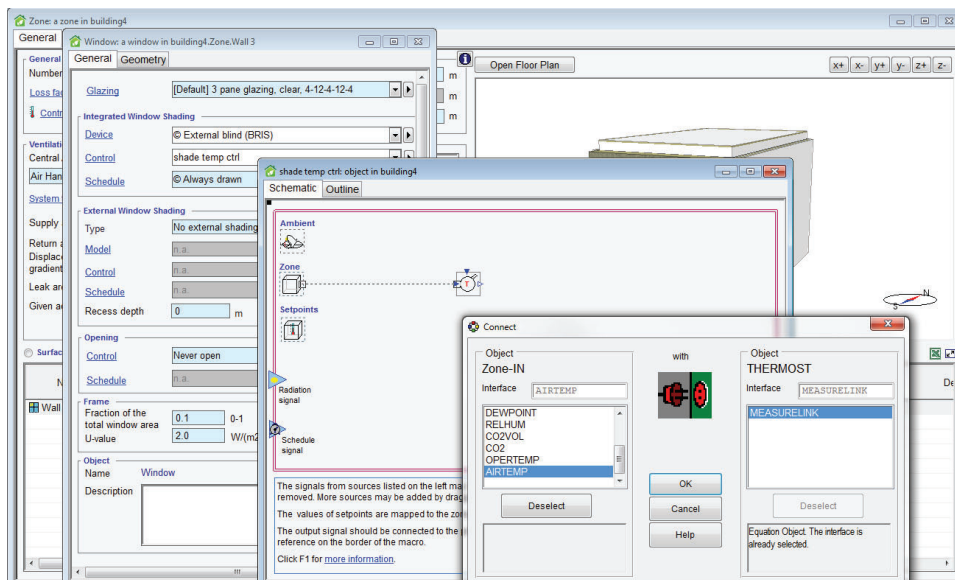


Figure 5.4

To achieve the right performance of the thermostat we have to give a dead band with sign; in this case a negative sign to get out signal 1 for high measure signals and 0 for low. Finally we insert a Constant field from the Utility palette and connect it to the setpoint link of the thermostat. With a given setpoint value (22°C in Figure 5.5) the control system is defined. After the simulation the effect of the shade control can be checked in the diagram “Heat balance”.

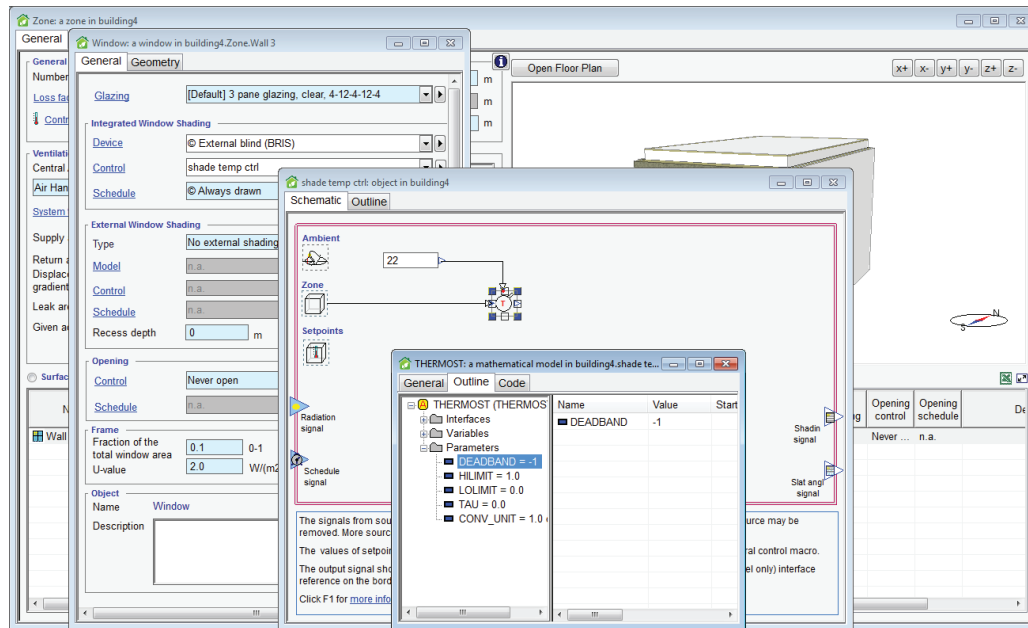


Figure 5.5

6. Tips and tricks

6.1. Speeding up computation

IDA ICE builds a single large simultaneous system of equations for all processes in the building. This system of equations contains several ordinary differential equations and has therefore different “built in” time constants. The room air will for example react quickly on a convective heat load, while the ground layer below the building can have a time constant several orders of magnitude larger. This system of equations is solved with numerical methods that adapt the timestep to the frequency content of the solution. Short time constants in the model in combination with high frequency content in driving functions (many starts and stops) can lead to long execution times.

Internally generated starts and stops (events) will also lead to short timesteps. Try for example to use a thermostat for radiator control.

The sheer number of equations in the system is naturally significant for the execution time, which increases roughly linearly with the problem size¹³. Therefore, it is vital not to model any unnecessary detail, and not for example model a large number of identical objects such as windows or cooling beams, when they could be replaced by a single larger one. Each added zone will contribute about two thousand variables to the system of equations.

The single most important factor for speeding up calculations is to have a reasonable amount of detail in time schedules (driving functions). Describing many sharp transitions will usually only result in increased computation time and not have a significant impact on results. From version 4, automatic smoothing is applied to key schedules to minimize these problems. (Can be turned off under System parameters).

For a model with reasonably simple time schedules, it can often prove effective to loosen the numerical tolerance to make the solver take longer and thereby fewer timesteps. Two important solver parameters are Tolerance and Maximum timestep. These are accessed from the Advanced tab of the Simulation data dialog. Often the tolerance can be relaxed to, say, 0.1-0.3 from the standard value 0.02. For problems with equations that are difficult to solve, it can sometimes be beneficial to instead decrease the tolerance. (The solver spends less time on failed attempts to take long steps.)

Looser tolerance will normally lead to acceptable loss of accuracy for accumulated quantities such as monthly energy consumption. For computing design (extreme values) of quantities such as heating or cooling load, one should be more careful with using loose tolerances and a large max timestep. Too loose a tolerance will on the other hand lead to decreased robustness, forcing the solver to often have to back up and retry with a smaller timestep, ultimately leading again to even longer execution time (if indeed the simulation is successful).

¹³ For more primitive numerical methods, the execution time will typically grow as the cube of the problem size.

To learn about the statistics of a simulation in terms of number of timesteps, variables, restarts etc. view the file screen.txt in the IDA temporary directory idamod48¹⁴. At the end of this file some statistics are given.

6.2. Time-split Parallelization

The clock time of the simulation can be substantially decreased on a computer with more than one core, able to run concurrent processes by splitting the whole simulation period into smaller periods running on different cores. This is implemented in IDA ICE to be done automatically if the box “Time-split parallelization” in the Advanced Simulation Data tab is ticked, see Figure 6.1. The number of processes is set under the Simulation Preferences menu, see Figure 6.2.

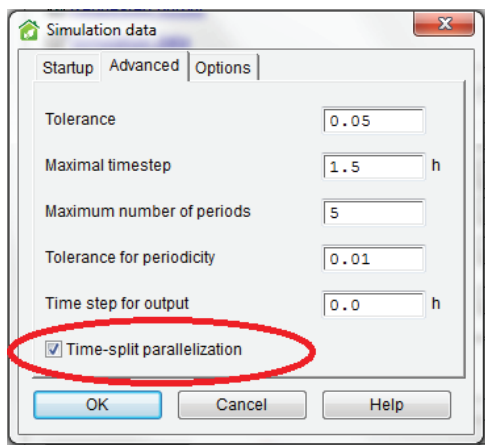


Figure 6.1

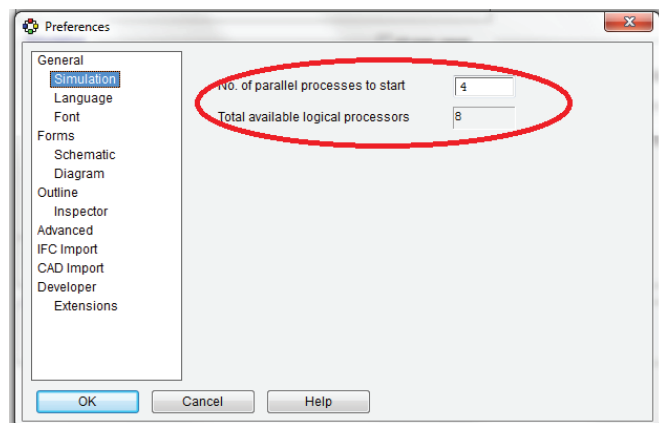


Figure 6.2

There is no gain in simulation time if the number of processes is higher than the total available logical processors and this parallelization technique has some overhead which makes it beneficial only for long simulation periods. The Speed-up is roughly 2-3 using 4 processes and roughly 4 using 12 processes.

6.3. Numerical instabilities

With a tool like *IDA ICE* it is easy to build large non-linear systems of equations and solve them for thousands of time points. However, it is impossible to, even theoretically, guarantee the success of the solution procedure. Any non-linear system of equations may have more than a single solution or none at all. Numerical computer programs are in this respect different from most other types of software, where it may be, at least theoretically, possible to create a bug-free code. *IDA ICE* also has a more difficult task than most other building simulation software, where less freedom is given to create mathematically complex models.

A major part of the IDA development work is devoted to improving the solver performance on difficult cases. However, this work is altogether dependent on close interaction with users. It is vital that users, which have built reasonable and meaningful models that are difficult to

¹⁴ This directory may occur in different locations depending on Windows version used. The path to the temporary directory can be found under Options > Preferences > Advanced. Solver files can also be viewed from the View menu > Solver files.

solve, take the trouble of sending the model to the support office. This is most easily done by using the Mail support function on the Help menu.

Some physical processes more often lead to problematic models. Generally speaking, models with significant natural ventilation flows through openings or leaks are the most difficult to solve, especially when the effect of wind pressure is included. If, in addition, vertical temperature gradients are to be simultaneously computed, one obtains a severely stiff and non-linear system of equations.

Avoid to use vertical or horizontal openings unless the bidirectional flow is essential for the study (use large leaks instead). Absolutely do not use large openings for the purpose of recreating the exact geometry of a real building. Normally, precise geometry has a very small impact on results.

A common type of error is when the user has graphically defined a piecewise linear controller and unintentionally entered several points near each other, giving the curve (in micro scale) several sharp corners and jumps. The solver will invariably have problems to negotiate the sharp turns of such a graph. Check the table view of such a curve for unintentional points.

A general way of dealing with difficult cases is to decrease the tolerance parameter that was discussed in the previous section. This forces the solver to be more careful and take smaller steps, which in most cases improves robustness. A tolerance of 0.001 or even smaller can sometimes be used.

A frequent situation is that any change in input data makes a previously failed run go through. This is not as strange as it sounds, since each change will lead to a different sequence of timesteps and in this way the exact combination of values that led to the failed timestep is avoided.

Another often effective trick is to replace sharp steps in schedules with steep ramps, especially for the fan control schedule. This will enable the solver to gradually over a short time period approach the new solution and thereby reach it more securely. If, in addition, the start and endpoints of the ramp are marked as input events (double entries of the same point in the table view), the solver will be even more cautious. Input events are marked by repeating the same time point twice in the profile, something best done in the Data tab view.

From version 4.7, a new useful diagnostic tool has been added. When the option ‘Show slowest model’ (Options menu, Preferences, Developer) has been selected, the component with the slowest convergence in each timestep will blink red during simulation, when viewed on the Schematic tab. For a case that runs well, the role of being the slowest model will change among several components during the simulation. On the other hand, for models with some particularly problematic equations, the components that are involved in this difficulty will blink more than others and this will frequently lead the user to understand the root of the problem.